CHE 555 – Homework 1

Full Solution

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  PROBLEM 1
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  SETUP
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  Curve: y = 2x² + 3x + 1
  The distance from a point (x, y(x)) to the origin is
      D(x) = sqrt[ x² + y(x)² ]
  Minimising D² :
      D²(x) = x² + (2x² + 3x + 1)²
      dD²(x*)/dx = 2x + 2(2x² + 3x + 1)(4x + 3) = 0
      2x + 2(2x² + 3x + 1)(4x + 3) = 0 -> 8x³ + 18x² + 14x + 3 = 0
      The roots of this cubic equations were solved numerically using the Newton-Raphson method.
  ────
  Optimal x      = -0.34140866
  y(x*)          = 2(-0.341409)² + 3(-0.341409) + 1
                 = 0.20889376
  D²(x*)         = (-0.341409)² + (0.208894)²
                 = 0.16019648
  D(x*)          = √0.16019648 = 0.40024552

  ANSWER
  ──────
  Nearest point : (-0.341409,  0.208894)
  Distance      : 0.400246
  error estimate : 1.94e-05
  iterations     : 18

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  PROBLEM 2
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  Variables : x = side length of square base (cm)
              h = height (cm)

  Volume constraint : x² · h = 1000  ->  h = 1000 / x²

  Surface area (no top lid) :
      S(x) = x²  +  4·x·h
           = x²  +  4·x·(1000/x²)
           = x²  +  4000/x

  ANALYTICAL SOLUTION
  ───────────────────
  dS/dx = 2x − 4000/x²  = 0
```

$2x^3 = 4000$
$x^3 = 2000$
$x^* = 2000^{(1/3)}$

$d^2S/dx^2 = 2 + 8000/x^3 > 0$ for all $x > 0$

NUMERICAL VALUES

$x^* = 2000^{(1/3)} = 12.599210$ cm
$h^* = 1000 / x^{*2} = 1000 / 12.599210^2 = 6.299605$ cm
$S^* = (12.5992)^2 + 4000/12.5992 = 476.220316$ cm^2
$d^2S/dx^2|x^* = 6.0000 > 0$

ANSWER

Optimal base side length : 12.5992 cm
Optimal height : 6.2996 cm
Minimum surface area : 476.2203 cm^2

PROBLEM 3

Minimise: $f(x_1, x_2) = 4x_1 - x_1^2 - 12$

Subject to:
  $g_1 : 25 - x_1^2 - x_2^2 = 0$
  $g_2 : 10x_1 - x_1^2 + 10x_2 - x_2^2 - 34 \geq 0$
  $g_3 : (x_1-3)^2 + (x_2-1)^2 \geq 0$
  $x_1, x_2 \geq 0$

VARIABLE COUNT

Total Variables : 2 (x_1, x_2)

DOF ANALYSIS

Each independent equality constraint reduces DOF by 1.
Equality constraints : 1
DOF = total variables − equality constraints = 2 − 1 = 1

Number of independent variables : 1
One valid choice : { x_1 }

EXPLANATION

E1 defines x_2 as a function of x_1: $x_2 = \sqrt{25 - x_1^2}$ (with $x_2 \geq 0$).
Once x_1 is chosen (satisfying $0 \leq x_1 \leq 5$), x_2 is fully determined.
The inequality constraints further restrict the feasible range of x_1
but do not reduce the number of independent variables.
$g_3 = (x_1-3)^2 + (x_2-1)^2 \geq 0$ is always satisfied (sum of squares).

ANSWER

Total variables : 2
Independent variables : 1
One possible choice : { x_1 }

PROBLEM 4

GIVEN

Cost per barrel : $C(P) = 50 + 0.1P + 9000/P$ (\$/barrel)
Selling price : \$300 / barrel
P = production rate (barrels/day)

PART A – Minimise cost per barrel

```
dC/dP = 0.1 − 9000/P²  = 0
P²  = 9000 / 0.1 = 90000
P*  = sqrt(90000) = 300 barrels/day

d²C/dP² = 18000/P³  > 0  for P > 0

Numerical check:
  P*     = 300.0 barrels/day
  C(P*)  = 50 + 0.1(300.0) + 9000/300.0 = 110.00 $/barrel
  d²C/dP²|P* = 0.000667 > 0
```

PART B – Maximise daily profit

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Revenue per day  : R(P) = 300 · P
Total cost / day : TC(P) = C(P) · P = 50P + 0.1P² + 9000
Profit / day     : Π(P) = R − TC = 300P − 50P − 0.1P² − 9000
                         = 250P − 0.1P² − 9000

dΠ/dP = 250 − 0.2P  = 0
P*  = 250 / 0.2 = 1250 barrels/day

d²Π/dP² = −0.2  < 0

Numerical check:
  P*           = 1250 barrels/day
  Revenue      = 300 × 1250             = $375,000.00/day
  Total cost   = 50(1250) + 0.1(1250)² + 9000 = $227,750.00/day
  Profit Π(P*) = $375,000.00 − $227,750.00   = $147,250.00/day
```

ANSWERS

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(A) Production minimising cost/barrel : P = 300 barrel/day  -> C = $110.00/barrel
(B) Production maximising daily profit: P = 1250 barrel/day  -> Π = $147,250.00/day
```

PROBLEM 5

DATA

x	y
10	1.00
20	1.26
30	1.86
40	3.31
50	7.08

MODELS TO FIT

```
Model 1 : y = exp(a + b·x)            [exp-linear]
Model 2 : y = exp(a + b·x + c·x²)     [exp-quadratic]
Model 3 : y = a · x^b                 [power law]
```

METHOD

Models 1 & 2 are linearised by taking ln y and fitting via OLS.
Model 3 is linearised as ln y = ln a + b·ln x and fitted via OLS.
Full regression statistics (R^2, adj-R^2, SE, RMSE, F-test, t-tests, ANOVA table) are reported for each model in the sections below.

Regression details:

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────────────────────────────────────────────────────────────
Model: y = exp(a + b·x)
────────────────────────────────────────────────────────────
Observations     : 5
Parameters (p)   : 2
DOF (error)      : 3

── Goodness of Fit ──────────────────────────────
R²               : 0.955527
R² (adjusted)    : 0.940703
SE of regression : 0.192228
RMSE             : 0.148900
SSE              : 0.110855
SSR              : 2.381815
SST              : 2.492670

── Overall F-Test ───────────────────────────────
F-statistic      : 64.4574
F p-value        : 0.00403481

── Parameter t-Tests  (H₀: param = 0, α = 0.05) ─
Param        Estimate      Std Err     t-stat      p-value   95% CI lower   95% CI upper
────────────────────────────────────────────────────────────────────────────────────
a           -0.662933     0.201611    -3.2882     0.046141      -1.304549      -0.021317  *
b            0.048804     0.006079     8.0285   0.00403481       0.029458       0.068149  *
(* p < 0.05)

── ANOVA Table ──────────────────────────────────
Source           df               SS               MS            F      p-value
──────────────────────────────────────────────────────────────────
Regression        1         2.381815         2.381815      64.4574   0.00403481
Residual          3         0.110855         0.036952
Total             4         2.492670

────────────────────────────────────────────────────────────
Model: y = exp(a + b·x + c·x²)
────────────────────────────────────────────────────────────
Observations     : 5
Parameters (p)   : 3
DOF (error)      : 2

── Goodness of Fit ──────────────────────────────
R²               : 0.999968
R² (adjusted)    : 0.999936
SE of regression : 0.006314
RMSE             : 0.003993
SSE              : 0.000080
SSR              : 2.492591
SST              : 2.492670

── Overall F-Test ───────────────────────────────
F-statistic      : 31265.9950
F p-value        : 3.19826e-05

── Parameter t-Tests  (H₀: param = 0, α = 0.05) ─
Param        Estimate      Std Err     t-stat      p-value   95% CI lower   95% CI upper
────────────────────────────────────────────────────────────────────────────────────
a           -0.040266     0.013541    -2.9736     0.096929      -0.098528       0.017997
b           -0.004568     0.001032    -4.4264    0.0474371      -0.009008      -0.000128  *
c            0.000890     0.000017    52.7166  0.000359642       0.000817       0.000962  *
(* p < 0.05)

── ANOVA Table ──────────────────────────────────
Source           df               SS               MS            F      p-value
──────────────────────────────────────────────────────────────────
Regression        2         2.492591         1.246295   31265.9950  3.19826e-05
Residual          2         0.000080         0.000040
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  Total             4         2.492670

────────────────────────────────────────────────────────────────────────────────
  Model: y = a·x^b [log-log space]
────────────────────────────────────────────────────────────────────────────────
  Observations     : 5
  Parameters (p)   : 2
  DOF (error)      : 3

  ── Goodness of Fit ──────────────────────────────
  R²               : 0.818733
  R² (adjusted)    : 0.758311
  SE of regression : 0.388089
  RMSE             : 0.300612
  SSE              : 0.451838
  SSR              : 2.040832
  SST              : 2.492670

  ── Overall F-Test ───────────────────────────────
  F-statistic      : 13.5502
  F p-value        : 0.0347317

  ── Parameter t-Tests  (H₀: param = 0, α = 0.05) ─
  Param        Estimate      Std Err     t-stat      p-value   95% CI lower   95% CI upper
  ──────────────────────────────────────────────────────────────────────────────────────
  ln a        -2.863028     1.010440    -2.8334    0.0660019      -6.078700       0.352644
  b            1.123962     0.305337     3.6811    0.0347317       0.152245       2.095680  *
  (* p < 0.05)

  ── ANOVA Table ──────────────────────────────────
  Source           df               SS               MS            F      p-value
  ──────────────────────────────────────────────────────────────────
  Regression        1         2.040832         2.040832      13.5502    0.0347317
  Residual          3         0.451838         0.150613
  Total             4         2.492670

Original-scale R² for Model: y = a·x^b: 0.726703  (adj R²: 0.635604)

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  MODEL COMPARISON SUMMARY
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  Model                        R²     adj R²         RMSE       F-stat      F p-val
  ────────────────────────────────────────────────────────────────────────────
  exp_linear             0.955527   0.940703     0.148900      64.4574   0.00403481
  exp_quadratic          0.999968   0.999936     0.003993   31265.9950  3.19826e-05
  power_law              0.726703   0.635604     0.300612      13.5502    0.0347317

  Best model (highest adj R²): exp_quadratic

  FITTED VALUES vs. DATA
  ─────────────────────────────────────────────────────────────────
      x     y_obs     exp-lin    exp-quad       power
  ────    ──────    ────────    ────────    ────────
     10    1.0000      0.8395      1.0030      0.7596
     20    1.2600      1.3677      1.2513      1.6554
     30    1.8600      2.2282      1.8650      2.6111
     40    3.3100      3.6300      3.3210      3.6079
     50    7.0800      5.9136      7.0651      4.6364

  RESIDUALS
  ─────────────────────────────────────────────────────────────────
      x     exp-lin    exp-quad       power
  ────    ────────    ────────    ────────
     10      0.1605     -0.0030      0.2404
     20     -0.1077      0.0087     -0.3954
     30     -0.3682     -0.0050     -0.7511
     40     -0.3200     -0.0110     -0.2979
     50      1.1664      0.0149      2.4436
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  CONCLUSION
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  The exp-quadratic model has the highest adjusted R²
  among the three candidates.
  Best model: exp-quadratic

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  END OF REPORT
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Plot

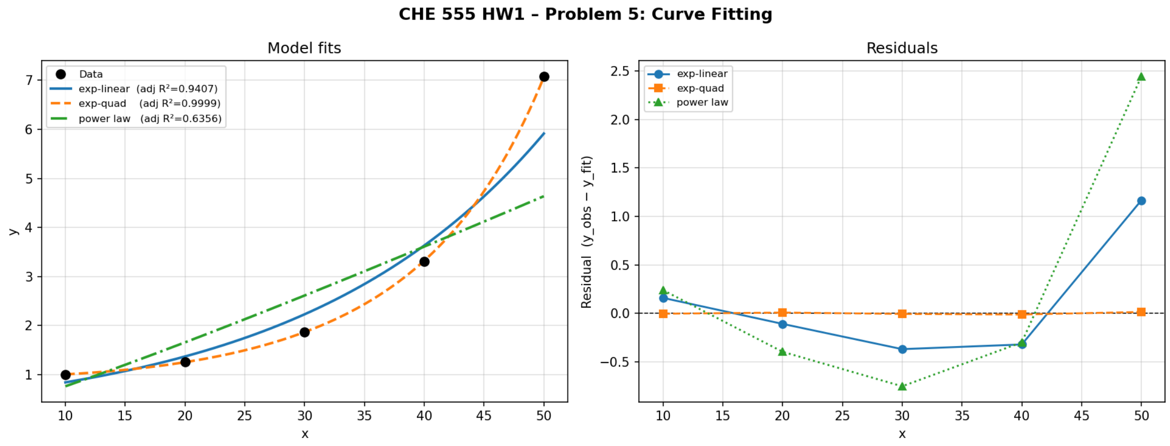

Figure 1: Curve fitting comparison for Problem 5.